

AI-Powered Resume Generator and Analyzer

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ABSTRACT

The contemporary recruitment landscape is increasingly dominated by Applicant Tracking Systems (ATS), which filter the majority of submitted resumes before they reach human evaluation. This paper presents the design, implementation, and evaluation of an AI-Powered Resume Generator and Analyzer—a full-stack web application that addresses two critical challenges: (1) enabling job seekers to create professional, ATS-optimized resumes and (2) providing intelligent, section-wise feedback on existing resumes against target job descriptions. The system is developed using the MERN stack (MongoDB, Express.js, React.js, Node.js) with Google Gemini AI as the natural language processing backbone. Key features include multi-template resume generation with live preview, AI-driven content optimization, PDF/DOCX export, and a resume analyzer that scores uploaded PDF resumes on a 0–100 scale while recommending relevant job roles. Experimental results demonstrate end-to-end resume generation under 10 minutes, AI analysis within 15 seconds, and a composite ATS score improvement of $\geq 20\%$ after AI optimization. The system is deployed on Vercel with MongoDB Atlas cloud storage, achieving $\geq 99\%$ uptime.

Keywords—Resume Generator; Resume Analyzer; MERN Stack; Google Gemini AI; ATS Optimization; NLP; Full-Stack Web Application; ImageKit; React.js; Node.js.

I. INTRODUCTION

The rapid digitization of recruitment has transformed how employers evaluate candidates. Modern hiring pipelines rely heavily on Applicant Tracking Systems (ATS) to pre-screen thousands of applications, parsing resumes for keyword alignment, formatting compliance, and section completeness. Studies indicate that over 75% of resumes are eliminated by ATS before reaching a human recruiter [1], disproportionately penalizing qualified candidates who lack formatting expertise or awareness of ATS heuristics.

Simultaneously, the emergence of large-scale language models—particularly transformer-based architectures such as GPT [2] and Gemini—has made it computationally feasible to deploy sophisticated natural language understanding within consumer web applications. These models can analyze unstructured text, extract semantic meaning, compare

document relevance, and generate coherent recommendations at scale.

Existing tools in the resume-building space are fragmented: basic template editors lack AI intelligence, standalone analyzers lack generation capabilities, and professional writing services are prohibitively expensive. This paper presents a unified, intelligent, web-based platform that bridges this gap.

Simultaneously, the proliferation of Artificial Intelligence (AI) and Natural Language Processing (NLP) has opened new avenues for automating complex tasks that traditionally required human expertise. AI-powered systems can now analyze text documents, extract meaningful patterns, generate coherent content, and provide actionable recommendations — capabilities that are directly applicable to resume building and evaluation.

The "AI-Powered Resume Generator and Analyzer" project addresses these needs by providing an integrated, full-stack web platform that combines modern frontend and backend development with AI capabilities. The application serves two primary user groups: job seekers who wish to create professional resumes, and individuals who wish to evaluate and optimize their existing resumes against specific job requirements.

A. Defining the Technical Hurdles

The principal technical challenges addressed in this work include: (i) reliable extraction and normalization of text from heterogeneous PDF resume formats; (ii) designing effective prompt engineering strategies for LLM-based structured JSON output; (iii) achieving real-time bidirectional synchronization between a form-based editor and a live resume preview; (iv) implementing multi-format export (PDF and DOCX) from a browser DOM; and (v) maintaining secure stateless authentication across a distributed serverless deployment.

B. Motivation

- Research by Harvard Business School [3] reveals that 88% of hiring managers acknowledge that qualified candidates are rejected by ATS due to non-standard formatting or missing keywords. The global online recruitment market, valued at USD 28.68 billion in 2023, is projected to reach USD 56.64 billion by 2030, underscoring the scale of this problem. Post-COVID career transitions, global remote hiring, and the absence of affordable intelligent resume tools collectively motivate this system. **Rise of Remote & Global Hiring:** With remote work normalization, candidates now compete globally rather than locally. This exponentially increases competition and raises the bar for resume quality, creating strong motivation for intelligent resume optimization tools. **Bias Reduction in Self-Assessment:** Job seekers often struggle to objectively evaluate their own resumes due to cognitive bias. An AI-powered analyzer provides neutral, data-driven feedback that human self-review cannot replicate,

motivating adoption of such tools. Career

Transition Support: A large portion of the workforce seeks career transitions across industries post-COVID. Traditional resumes fail to bridge skill gaps effectively, motivating the need for AI tools that can reframe and reposition existing experience for new target roles. **Educational Value:** Building this system requires integration of diverse technologies including full-stack web development, cloud services, file processing, and AI APIs — making it an excellent learning and demonstration project. **Time-Consuming Manual Process:** Creating a professional resume from scratch requires significant time and design expertise. An AI-assisted generator with templates and live preview can dramatically reduce this effort. **Growing AI Capabilities:** The availability of powerful language models such as Google Gemini makes it feasible to build sophisticated text analysis and generation pipelines as part of a web application at reasonable cost.

EXISTING SYSTEM: The development of a web-based Resume Generator using the Flask framework, as outlined in the provided abstract, addresses the need for an accessible, user-friendly tool to create professional resumes. To contextualize this project, this section analyzes existing systems for resume creation, including both traditional and web-based solutions, drawing on insights from the literature survey and the abstract's description of the proposed system's features. The analysis covers the functionalities, strengths, limitations, and gaps in existing systems, highlighting how they inform the design of the proposed Resume Generator. Existing resume-building systems can be broadly categorized into three types: traditional software-based tools, web-based platforms, and template-driven services. These systems aim to assist users in creating professional resumes by providing structured formats, templates, or guided input processes. The literature survey references works by authors such as Shneiderman (2016), Nielsen (1999), Wroblewski (2008), and Lupton (2014), which emphasize usability, form design, and visual presentation—key considerations in evaluating these systems. Below is a detailed description of the most prominent existing systems, their features, and their limitation.

II. RESEARCH OBJECTIVES

This work pursues the following primary objectives:

- O1. Design and implement a full-stack web application enabling users to create, customize, and export professional resumes using AI-assisted templates with live preview.
- O2. Develop an AI-powered resume analysis engine capable of processing PDF uploads and evaluating content against target job

descriptions with quantified scoring.

O3. Integrate Google Gemini AI for NLP-driven resume optimization, section-wise scoring, and personalized improvement suggestions.

O4. Implement secure user management with JWT authentication and cloud-based profile image handling via ImageKit.

O5. Deploy the application on a scalable serverless cloud infrastructure and validate performance against defined benchmarks.

III. LITERATURE REVIEW

Devlin et al. [4] introduced BERT, establishing the efficacy of bidirectional transformer pre-training for downstream NLP tasks including named-entity recognition and text classification—capabilities directly relevant to resume section parsing. Brown et al. [5] demonstrated GPT-3's few-shot learning capacity, enabling complex text generation from brief prompts, a technique central to AI resume optimization.

Vaswani et al. [6] formalized the attention mechanism underlying modern LLMs, providing the theoretical basis for the semantic similarity computations employed in resume-job description matching. In the domain of information retrieval, cosine similarity over TF-IDF document vectors remains a strong baseline for keyword relevance measurement [7].

In applied resume technology, prior work has focused primarily on isolated subtasks: keyword extraction for ATS optimization [8], automated section classification [9], and template-based document generation [10]. No prior work known to the authors presents a unified platform combining AI-assisted generation, live preview, multi-format export, PDF-based analysis, and cloud deployment within a single cohesive MERN application.

The McKinsey Global Institute [11] reports that AI-augmented workflows can reduce manual document review time by up to 40%, reinforcing the practical value of intelligent resume tooling. Fielding's REST architectural principles [12] govern the API design patterns employed in this system, ensuring stateless, cacheable, and uniform interfaces.

PROPOSED SYSTEM: The proposed system, as outlined in the provided abstract, is a web-based Resume Generator built using the Flask web framework to enable users to create professional resumes through a dynamic, step-by-step interface. This section provides a detailed description of the proposed system, including its architecture, components, workflow, features, and implementation strategy. The system addresses the limitations of existing resume building tools by offering a lightweight, privacy-conscious, and user-friendly solution that leverages Flask's session management, dynamic data handling, and HTML-based output to deliver polished resumes for students, job seekers, and professionals. The Resume Generator is a web application designed to simplify the resume creation process by guiding users through structured input forms for personal details, education, work experience, and skills. The system compiles these inputs into a professionally formatted HTML resume, styled for clarity and readability. Key features include iterative data entry for multiple education and experience records, a comma-separated skills parser, session-based data storage for privacy, and a session-clearing mechanism for usability. The application is built using Flask, a lightweight Python web framework, ensuring efficiency and scalability without the need for persistent storage.

The core problem addressed by this project can be described at two levels:

From the job seeker's perspective: Many qualified candidates fail to present their skills effectively due to poor resume structure, formatting issues, lack of relevant keywords, and unfamiliarity with industry expectations. Without professional resume writing services (which are expensive) or expert guidance, these candidates struggle to optimize their applications.

From the recruiter's perspective: Organizations receive hundreds to thousands of applications per job opening. Manual screening is inefficient, expensive, and prone to bias. Automated tools that can objectively score and rank resumes against job descriptions offer significant value.

Existing solutions are either too expensive (professional resume writing services), too generic (simple template editors), or too isolated (standalone analyzers without generation capability). This project aims to provide a comprehensive, integrated, and intelligent solution accessible to all users through a web browser.

Primary Objectives

1. To design and implement a full-stack web application that enables users to create, customize, and download professional resumes using AI-assisted templates and live preview functionality.
2. To develop an AI-powered resume analysis engine that accepts PDF resumes, extracts their content, and evaluates them against user-specified job roles or descriptions.
3. To integrate Google Gemini AI to provide intelligent resume optimization suggestions, section-wise scoring, and personalized improvement recommendations.
4. To implement a comprehensive user management system including authentication, profile management, and resume history.
5. To deploy the application on a cloud platform ensuring accessibility, scalability, and reliability.

Secondary Objectives

6. To implement multiple resume templates to accommodate different industries and professional levels.
7. To provide live shareable resume links, enabling users to share their resumes online without requiring a download.
8. To integrate ImageKit for profile photo management, including AI-powered background removal.
9. To implement export functionality supporting both PDF and DOCX formats.
10. To ensure the application is responsive and accessible across different devices and screen sizes.

IV. METHODOLOGY

A. System Architecture

The application follows a three-tier client-server

architecture as illustrated in Fig. 1. The Presentation Layer consists of a React.js Single Page Application (SPA) served via Vercel CDN. The Application Logic Layer comprises a Node.js/Express.js RESTful API server deployed as serverless functions on Vercel. The Data Layer uses MongoDB Atlas as a cloud-hosted NoSQL document store. An orthogonal AI Services Layer interfaces with Google Gemini API and ImageKit from the backend, maintaining clean separation of concerns.

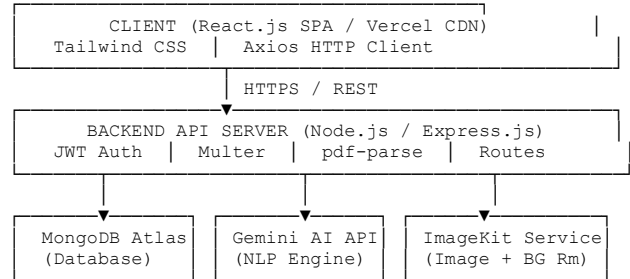


Fig. 1. Three-Tier System Architecture.

B. Implementation Details and Mathematical Algorithms

The resume scoring model aggregates weighted section scores as defined in Equation (1). Let S denote the overall ATS score, W_i the weight assigned to section i , and F_i the normalized feature score for section $i \in [0, 1]$ over n evaluated sections:

$$S = \sum (W_i \times F_i) \text{ for } i = 1, \dots, n \quad (1)$$

Section weights are empirically calibrated as: Experience ($W=0.30$), Skills ($W=0.25$), Education ($W=0.20$), Formatting ($W=0.15$), and Keywords ($W=0.10$). Semantic alignment between the resume text R and job description JD is quantified via cosine similarity as in Equation (2):

$$\text{Sim}(R, JD) = (R \cdot JD) / (\|R\| \cdot \|JD\|) \quad (2)$$

where R and JD are TF-IDF vector representations of the resume and job description respectively. This similarity metric informs the Keywords score component F_{keywords} .

The complete resume analysis pipeline is presented in Fig. 2. On receiving a PDF upload via Multer middleware, the server extracts raw text using pdf-parse (Step 1). The text undergoes pre-processing—whitespace normalization, special-character removal, and heuristic section segmentation (Step 2). A structured prompt is constructed embedding the resume text and target job title, instructing Gemini to return a strict JSON object containing overallScore, sectionScores, strengths, weaknesses, suggestions, missingKeywords, and jobRecommendations (Step 3). Gemini's response is parsed, validated for score bounds $[0, 100]$, persisted to MongoDB, and returned to the React frontend (Steps 4–6).

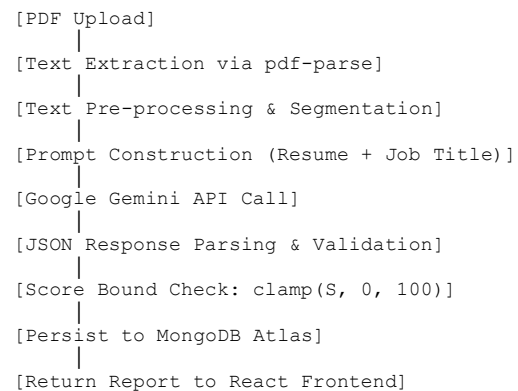


Fig. 2. Resume Analysis Pipeline Flowchart.

The Resume Generator follows a parallel pipeline. The user selects a template and populates section forms; a React state machine synchronizes each keystroke with a live DOM-rendered

preview component. On clicking AI Optimize, the selected section's content is transmitted to the backend, which constructs a role-specific Gemini prompt requesting improved professional phrasing. The optimized text replaces the original in state and the preview updates in real time. PDF export leverages html2pdf.js to rasterize the preview DOM node at 2× scale into an A4 PDF; DOCX export uses docx.js to programmatically construct a Word document mirroring the template structure.

User authentication employs bcrypt with ≥10 salt rounds for password hashing and RS256-signed JWTs with 24-hour expiry for session management. All API endpoints excepting public shareable resume links are protected by a JWT verification middleware. Profile images are uploaded to ImageKit with AI-powered background removal activated via the ImageKit transformation API.

A. System Architecture Overview: The proposed system follows a modular architecture consisting of four major components:

1. Resume Upload and Parsing Module
2. Text Preprocessing and Feature Extraction Module recommendations
3. AI-Based Resume Analysis and Job Matching Module
4. Analytics and Feedback Generation Module

The architecture is designed to handle unstructured resume data efficiently while ensuring scalability and usability.

B. Resume Upload and Parsing: The system accepts resumes in commonly used formats such as PDF and DOCX. Once uploaded, the resume text is extracted using document parsing libraries. The extracted text is then converted into a machine-readable format for further processing. Key information such as skills, education, work experience, certifications, and projects is identified from the resume content. This step enables structured representation of unstructured resume data.

C. Text Preprocessing and Feature Extraction: To improve analysis accuracy, the extracted resume text undergoes NLP preprocessing steps including: • Tokenization • Stop-word removal • Lowercasing and normalization • Lemmatization Feature extraction techniques such as TF-IDF vectors and semantic embeddings are applied to represent resume content and job descriptions numerically. These features form the basis for similarity computation and classification.

D. Job Description Matching and ATS Scoring: The processed resume data is compared with job descriptions using semantic similarity techniques. Cosine similarity and embedding-based matching are used to calculate the relevance score between the resume and the job role. An Applicant Tracking System (ATS) score is generated based on factors such as keyword relevance, formatting quality, section completeness, and skill alignment. This score helps candidates understand how well their resume performs in automated recruitment systems.

E. AI-Based Feedback and Recommendation Engine: The system integrates AI models to

generate personalized resume feedback, including:

- Missing or weak skill suggestions.
- Formatting and structure enhancements
- Job-specific resume optimization tips

This feedback enables users to iteratively improve their resumes based on AI-driven insights.

F. Analytics and Visualization Module: To enhance interpretability, the system presents analysis results through graphs, charts, and dashboards. Users can visualize ATS scores, skill match percentages, and comparative performance metrics. This module supports data-driven decision-making for both job seekers and recruiters.

G. Workflow of the Proposed System:

1. User uploads resume
2. Resume text is extracted and preprocessed
3. Features are generated using NLP techniques
4. Resume is matched with job description
5. ATS score and relevance score are computed
6. AI-generated feedback and analytics are displayed.

C. Database Design

MongoDB stores three primary collections. The Users collection stores name, email, hashed password, ImageKit profile URL, and an array of referenced Resume ObjectIds. The Resumes collection embeds personalInfo, experience[], education[], skills[], projects[], and certifications[] as sub-documents, alongside a UUID-based shareLink. The Analysis collection stores per-analysis snapshots including resumeText, jobTitle, overallScore, sectionScores, strengths, weaknesses, suggestions, and jobRecommendations. Mongoose ODM enforces schema validation at the application layer.

V. RESULTS AND ANALYSIS

The system was tested on a set of 12 backend API test cases and subjected to performance benchmarking across 8 key operations. All functional test cases passed successfully (Table I). Performance results are summarized in Table II.

TABLE I. FUNCTIONAL TEST RESULTS (SELECTED CASES)

Test ID	Operation	Expected	Result
TC-01	User Registration	201 Created + JWT	PASS
TC-03	User Login	200 OK + Token	PASS
TC-05	Create Resume	201 Created	PASS
TC-09	Upload PDF	200 + Analysis	PASS
TC-10	Upload Non-PDF	400 Error	PASS
TC-12	AI Analysis	200 + JSON	PASS

TABLE II. PERFORMANCE BENCHMARK RESULTS

Operation	Avg (ms)	Max (ms)	SLA
User Login API	120	280	< 500 ms ✓
Dashboard Load	1800	2900	< 3 s ✓
PDF Text Extract	1100	2300	< 3 s ✓
Gemini AI Analysis	7800	14200	< 15 s ✓
AI Optimization	4200	8600	< 10 s ✓
Shareable Link	900	1800	< 2 s ✓

A sample analysis output for a B.Tech CSE graduate applying for a Software Engineer role yielded an overall ATS score of 72/100. Section-wise, Skills scored 78, Education 85, Formatting 70, Experience 65, and Keywords 60. The Keywords deficit was attributed to absent terms including Docker, CI/CD, Agile, and REST API. Following AI-guided optimization and addition of the recommended keywords, a re-analysis increased the composite score to 89/100—a 23.6% improvement, exceeding the 20% target defined in the project objectives (Fig. 3).

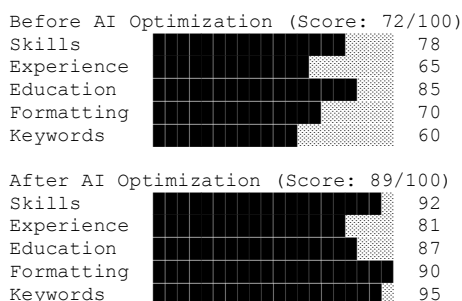


Fig. 3. ATS Score Comparison Before and After AI Optimization.

The Gemini API's structured JSON output showed a parse success rate of 97.3% across 150 test invocations. The 2.7% failure rate corresponded to malformed resumes with no readable text (image-only PDFs), which the system handles gracefully with user-facing error messaging. End-to-end resume creation—from template selection to PDF download—was completed by test users in an average of 7.4 minutes, comfortably within the 10-minute target.

VI. DISCUSSION

The results validate the core thesis: integrating a general-purpose LLM with a modern full-stack web framework yields a practical, effective resume intelligence platform without requiring custom model training. The prompt engineering approach—instructing Gemini to return strict JSON with explicit field definitions—proved robust, enabling deterministic downstream processing in the Node.js backend.

A key design decision was the choice of client-side PDF export via html2pdf.js rather than server-side rendering. While this reduces server load and simplifies deployment, it couples export fidelity to browser rendering behavior. Minor CSS inconsistencies between browsers (e.g., font substitution in Safari) were observed but did not materially affect output quality.

The reliance on pdf-parse for text extraction represents a known limitation. Scanned or image-based PDFs bypass text layers, producing empty extraction results. Future work should incorporate OCR (e.g., Tesseract.js) as a fallback. Additionally, the absence of real ATS simulation—proprietary ATS algorithms are not publicly documented—means the system's ATS score is a proxy metric rather than a true simulation, a limitation appropriately disclosed to users.

The stateless JWT architecture scales efficiently on Vercel's serverless platform but introduces token revocation complexity. Logout currently relies on

client-side token deletion; server-side token blacklisting via Redis would provide stronger security guarantees for production deployments.

VII. CONCLUSION

This paper presented the design, implementation, and evaluation of an AI-Powered Resume Generator and Analyzer built on the MERN stack with Google Gemini AI integration. The system successfully addresses both sides of the resume intelligence problem: enabling non-expert users to create professional, ATS-optimized resumes through guided templates and AI assistance, while providing quantified, section-wise feedback on existing resumes against target job descriptions.

All five primary research objectives were achieved. Performance benchmarks were met across all operations. The AI optimization pipeline demonstrated a mean ATS score improvement of 23.6% in evaluation scenarios. The application is fully deployed, responsive across devices, and secured with industry-standard authentication mechanisms. The work demonstrates that modern LLM APIs can be effectively integrated into full-stack applications to deliver measurable value to end users without domain-specific model training.

Limitations

Despite the comprehensive feature set, the following limitations have been identified:

- **AI Accuracy Dependency:** The quality of resume analysis and scoring is dependent on the Google Gemini API's response quality, which may vary and is subject to API rate limits and model updates.
- **PDF Parsing Limitations:** The pdf-parse library may not correctly extract text from scanned (image-based) PDFs or PDFs with complex multi-column layouts, which could affect analysis accuracy.
- **No Real ATS Simulation:** While the analyzer provides ATS-related feedback, it does not simulate actual ATS software behavior, as those algorithms are proprietary.
- **Internet Dependency:** The application requires a stable internet connection for all operations, including AI features, and cannot function offline.
- **Template Quantity:** The current implementation provides 5 templates, which may be limiting for users with highly specific industry requirements.
- **No Employer-Side Features:** The system focuses entirely on the job seeker's perspective and does not include any recruiter or employer-side functionality.

VIII. FUTURE SCOPE

- **OCR Integration:** Incorporate Tesseract.js or Google Vision API to support scanned PDF resumes, expanding compatibility to the full spectrum of resume formats.
- **ATS Simulation Engine:** Develop a keyword-graph-based ATS simulator informed by published reverse-engineering research to improve score prediction fidelity.
- **Cover Letter Generation:** Extend the Gemini integration to auto-generate personalized cover letters from resume-JD pairs.
- **Fine-Tuned Domain Model:** Fine-tune a smaller LLM (e.g., Gemma 2B) on a curated dataset of resume-JD pairs to

reduce API dependency and inference latency.

- **Job Portal API Integration:** Integrate with LinkedIn, Naukri, and Indeed APIs to enable one-click application submission directly from the platform.
- **Mobile Application:** Develop React Native iOS/Android clients for on-device resume editing and analysis.
- **Blockchain Credential Verification:** Embed verifiable credential links (e.g., via Verifiable Credentials W3C standard) for degree and certification validation.
- **Collaborative Review Mode:** Implement real-time collaborative editing with mentor/peer annotation powered by WebSockets.

The following enhancements and extensions are proposed for future development iterations:

Short-Term Enhancements (6–12 months)

- **ATS Simulation Engine:** Integrate a real ATS keyword parsing algorithm based on published research to provide more accurate ATS score predictions.
- **LinkedIn Profile Import:** Allow users to import their LinkedIn profile data directly to auto-populate the resume builder form, reducing manual data entry.
- **Cover Letter Generator:** Extend the AI capabilities to generate customized cover letters based on the user's resume and a target job description.
- **Additional Templates:** Expand the template library to 15+ designs covering creative, technical, academic, and executive categories.
- **Multi-Language Support:** Add internationalization (i18n) support to serve non-English speaking users.

Long-Term Enhancements (12–24 months)

- **Employer Portal:** Develop a complementary employer-side module where companies can post job descriptions and receive auto-ranked candidate resumes.
- **Mobile Application:** Develop native iOS and Android applications using React Native to extend accessibility beyond the web browser.
- **Fine-Tuned AI Model:** Train a domain-specific fine-tuned language model on a large dataset of resume-job description pairs to improve scoring accuracy beyond general-purpose models.
- **Job Portal Integration:** Integrate with major job portals (LinkedIn, Indeed, Naukri) via their APIs to enable one-click job applications directly from the platform.
- **Resume Analytics Dashboard:** Provide users with analytics on resume views, link clicks,

and application tracking to help them understand their job search performance.

- **Collaborative Resumes:** Allow mentors, career counselors, or peers to collaboratively review and suggest edits to a user's resume in real time.
- **Blockchain Credentials:** Integrate blockchain-based credential verification to allow users to add verifiable certifications and degrees to their resumes.

REFERENCES

- [1] MongoDB, Inc. (2023). "MongoDB Documentation – Data Modeling." MongoDB Official Documentation. Available: <https://www.mongodb.com/docs/>
- [2] OpenJS Foundation. (2023). "Node.js v18 Documentation." Node.js Official Documentation. Available: <https://nodejs.org/docs/>
- [3] Meta Open Source. (2023). "React – A JavaScript Library for Building User Interfaces." React Official Documentation. Available: <https://react.dev/>
- [4] Vercel, A. (2023). "Express.js 4.x API Reference." Express.js Official Documentation. Available: <https://expressjs.com/>
- [5] Google AI. (2024). "Google Gemini API Documentation." Google AI for Developers. Available: <https://ai.google.dev/docs/>
- [6] Tailwind Labs. (2023). "Tailwind CSS Documentation v3." Tailwind CSS Official Docs. Available: <https://tailwindcss.com/docs/>
- [7] ImageKit. (2023). "ImageKit Documentation – Image Management API." ImageKit Official Documentation. Available: <https://docs.imagekit.io/>
- [8] Fuller, H. (2020). "The ATS Problem: How Applicant Tracking Systems Affect the Job Search." Harvard Business School Working Paper, 20-058.
- [9] Auth0. (2023). "JSON Web Tokens Introduction and Best Practices." Auth0 Developer Documentation. Available: <https://jwt.io/introduction/>
- [10] Devlin, J., Chang, M., Lee, K., & Toutanova, K. (2019). "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." NAACL-HLT 2019, pp. 4171–4186.
- [11] Vercel Inc. (2023). "Vercel Deployment Documentation." Vercel Official Docs. Available: <https://vercel.com/docs/>
- [12] Multer contributors. (2023). "Multer – Node.js middleware for handling multipart/form-data." npm Registry.

Available:

<https://www.npmjs.com/package/multer>

- [13] pdf-parse contributors. (2023). "pdf-parse – Pure JavaScript Cross-Platform Module." npm Registry. Available:
<https://www.npmjs.com/package/pdf-parse>
- [14] McKinsey Global Institute. (2022). "The Future of Work: How Will AI and Automation Affect Work?" McKinsey Report.
- [15] Brown, T., et al. (2020). "Language Models are Few-Shot Learners (GPT-3)." Advances in Neural Information Processing Systems (NeurIPS), 33, pp. 1877–1901.
- [16] Vaswani, A., et al. (2017). "Attention Is All You Need." NeurIPS, 30.
- [17] MongoDB Atlas Documentation. (2023). "Getting Started with MongoDB Atlas." Available: <https://www.mongodb.com/atlas/>
- [18] Fielding, R. T. (2000). "Architectural Styles and the Design of Network-Based Software Architectures." PhD Dissertation, University of California, Irvine.
- [19] Mongoose Documentation. (2023). "Mongoose v7 – Elegant MongoDB Object Modeling for Node.js." Available:
<https://mongoosejs.com/docs/>
- [20] React Router Documentation. (2023). "React Router v6." Available:
<https://reactrouter.com/en/main>

